
Earning the Clinician's Trust: An Eight-Question Confidence Framework for Verified Physical AI in Oncology Trials

Kevin Kawchak 
CEO ChemicalQDevice
10.5281/zenodo.xxxxxxxx

June 15, 2026 (final-framework, v0.2.0)

Abstract.

This framework is written for the clinicians who will be asked to supervise autonomous Physical AI in oncology clinical trials, and who must decide, at the bedside, when that reliance is earned. It rests on a single mechanism, verification before generation, in which a software agent proposes a clinical action and a ten-gate verification, validation, and uncertainty quantification examination either accepts it under audit, escalates it to a qualified human, or blocks it before it can reach a patient. The decision to trust is organized as eight plain questions a clinician already asks of any tool: is it competent, is it safe, is it transparent, am I in control, is it equitable, is it reliable, who is accountable, and does it fit my workflow. Each question pairs a clinical concern with a credible, cited fact, and closes on the concrete mechanism, a gate, a record, a role, or a control, that answers it. The aim is not maximal trust but calibrated trust: reliance proportioned to demonstrated capability, avoiding both automation bias and algorithm aversion. The framework draws on a documented engineering lineage, including a surgical-humanoid assurance run that passed 172 of 172 automated tests across a ten-gate suite, and on the literature of trust in automation, clinical decision support, and algorithmic bias. Three questions recur throughout: would I let this system near my own family member, can I defend this decision at tumor board, and could I show an auditor exactly what happened and why. The conclusion is an adoption decision a clinician can sign.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords Physical AI | Oncology clinical trials | Verification before generation | VVUQ | Calibrated trust | Automation bias | Clinician adoption | Patient safety

1 Introduction: the questions a clinician asks at the bedside

This framework is written for a reader who treats cancer patients and who will be asked, soon, to supervise an autonomous system in a clinical trial, but who has never driven a surgical robot or written a line of

verification code. It defines its terms as it goes, and it organizes the decision to trust around eight plain questions a clinician already asks of any new tool.

A few definitions make the rest readable. *Physical AI* is artificial intelligence that acts in the physical world through a machine, such as a surgical humanoid in an oncology trial, rather than only returning text on a screen. *Verification before generation* is the principle that a machine may propose an action, but the proposal is checked against fixed, published standards before anything reaches a patient [1]. *Claude Code* is the software agent that drafts the candidate action, *Codex* is a second, independent agent that reviews it, and *VVUQ*, meaning verification, validation, and uncertainty quantification, is the ten-gate examination the proposal must pass. The examination resolves in exactly one of three ways: ACCEPT and act under audit, ESCALATE to a qualified human, or BLOCK before execution. Figure 1 shows the mechanism from the clinician’s seat.

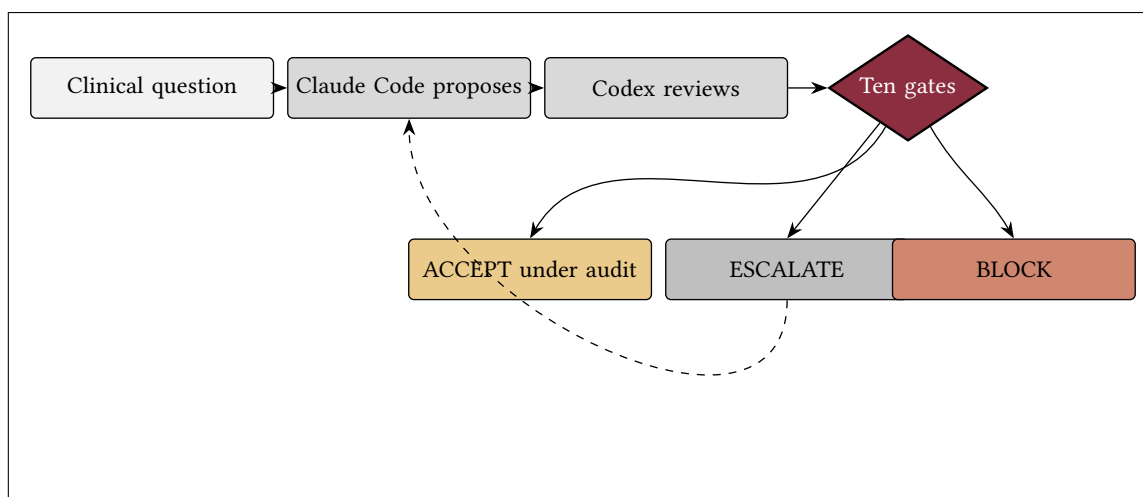


Figure 1. Verification before generation, the clinician’s view (Mermaid 03 reproduced as colored TikZ). A clinical question is posed, one agent proposes, a second reviews, and ten gates resolve to ACCEPT, ESCALATE, or BLOCK.

The goal of the framework is a particular kind of trust. *Calibrated trust* is reliance proportioned to a system’s demonstrated capability, no more and no less [2]. Its two failure modes are symmetric: *automation bias*, the over-trust that accepts a wrong output because a machine produced it [3], and *algorithm aversion*, the under-trust that discards a capable tool after seeing it err once [4]. The eight questions exist to calibrate, not to maximize. They are previewed in Figure 2.

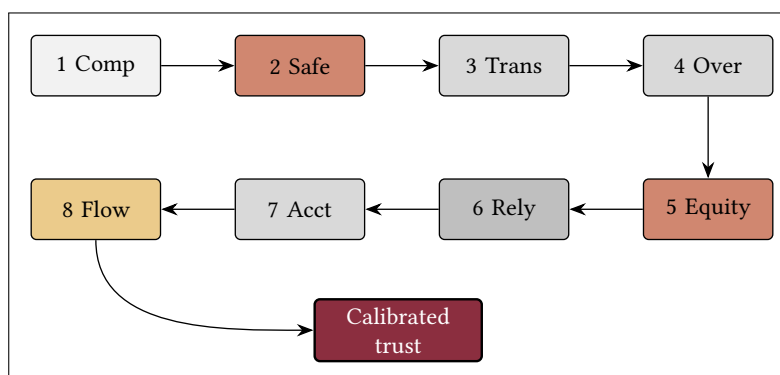


Figure 2. The eight confidence questions (Mermaid 02 reproduced as colored TikZ). The answers accumulate into calibrated trust.

The convergence of human and machine performance is now well documented in medicine [5], but documentation is not adoption. Three questions recur throughout, asked politely and answered concretely. Would I let this system near my own family member? Can I defend this decision at tumor board? And if an auditor asked me tomorrow, could I show exactly what happened and why? The remainder takes the eight confidence questions in turn, each defining its terms, pairing its claim with a cited fact, and closing on the specific mechanism that answers it.

Contents

1 Introduction: the questions a clinician asks at the bedside	1
2 Competence: is it actually good at this, or only good on paper?	4
3 Safety: what stops it before it hurts someone?	5
4 Transparency: can I see why it did that?	6
5 Oversight: am I still in control?	7
6 Equity: does it work for the patient in front of me?	9
7 Reliability: will it be the same tomorrow, at 3 a.m.?	10
8 Accountability: who is responsible when it is wrong?	11
9 Workflow: does it fit how I actually work?	13
10 Appendix A: figure compendium	16
References	19
Acknowledgments	22
Ethical Disclosures	23
Rights and Permissions	23
Cite This Framework	23
Data Availability	23

2 Competence: is it actually good at this, or only good on paper?

The first question a clinician asks of any tool is whether it is competent at the specific task, on patients like the ones in this clinic. Competence is not a single benchmark number; it is a stack of evidence built in order, from bench accuracy to prospective validation on the site's own population. The convergence of human judgment and machine pattern recognition only earns its place at the bedside when the machine has been shown to perform, not in the abstract, but on the question and the population in front of the clinician [5].

This is where the mechanism of the framework begins to matter. Every action the system contemplates is produced under verification before generation: Claude Code proposes a candidate step, Codex reviews that step independently, and a ten-gate VVUQ examination (verification that the action was built correctly, validation that it is the right action for this patient, and uncertainty quantification of how confident the system is) resolves the proposal to one of three outcomes: ACCEPT and act under audit, ESCALATE to a qualified human, or BLOCK before the patient. Competence, in this framing, is not a marketing claim but the documented record that this examination has been passed on evidence the clinician can inspect.

The author's assurance lineage shows what that record looks like in practice. A surgical-humanoid platform built for a mobile pancreatic-cancer unit was subjected to a priority VVUQ examination before any action was permitted [6], and the broader verification-automation work demonstrates how the same gate suite is applied to an oncology clinical trial rather than to a single procedure [7]. The headline number from the assurance run is concrete and checkable: the system passed 172 of 172 automated tests before it was cleared to act. A clinician does not have to take competence on faith; the clinician can ask to see which tests were run and how each resolved, which is the discipline the illustrative H. R. 9510 would write into the trial record [1].

The decisive adoption action here is local validation. A model that performs well on a vendor's bench, or even on a published retrospective cohort, has not yet been shown to perform on this site's patients, with this site's imaging protocols and case mix. Competence is therefore earned on a ladder, and each rung answers a different question while leaving others open. Table 1 sets out the ladder so the clinician can see precisely what a given level of evidence establishes and, just as important, what it cannot. The competence evidence stack in Figure 3 traces the same ascent, ending only when the assurance run closes the gap between paper performance and the population in the room.

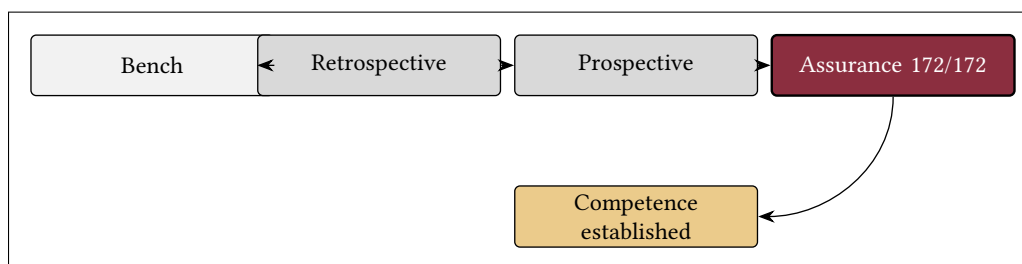


Figure 3. The competence evidence stack (Mermaid 06 reproduced as colored TikZ): bench accuracy, retrospective cohort, prospective on-site validation, and the assurance run advance in order until competence is established on the local population.

Evidence level	What it establishes	What it cannot show
Bench accuracy	The model meets a numeric performance threshold on a curated reference set	That it holds on real patients or this site's protocols
Retrospective cohort	Performance on historical records spanning a real population	That it acts safely in real time, before an outcome is known
Prospective on-site validation	Performance on this clinic's own patients under the actual workflow	That rare or catastrophic cases are handled, which require the gates
Assurance run (172 of 172 gate tests)	Every verification, validation, and uncertainty gate passed before action was permitted	A substitute for continued monitoring once the system is in routine use

Table 1: The competence evidence ladder: what each level of evidence establishes and what it leaves unanswered, culminating in local validation and the assurance run.

Competence is answered not by a claim but by evidence the clinician can inspect, and by the discipline of validating on the local population before clinical use. The honest test is the family test: would I let this system near my own family member on the strength of a bench number alone, or only after I had seen it validated on patients like the one in front of me and watched it pass every gate?

3 Safety: what stops it before it hurts someone?

The clinician's duty of non-maleficence makes safety the second question. It is not enough that a system usually behaves; the clinician needs to know what concretely stops an irreversible action before it can reach the patient. The fear behind this question is not abstract. The landmark report *To Err Is Human* estimated that tens of thousands of Americans die each year from preventable medical error [8], and a later analysis ranked medical error among the leading causes of death in the United States [9]. A safety case that deserves a clinician's trust must therefore name the specific thing that prevents the worst outcome, not merely promise that it is unlikely.

In this framework that specific thing is the ten-gate VVUQ examination. Every candidate action is generated under verification before generation, in which Claude Code proposes and Codex reviews independently, and then must pass ten gates before it may proceed. The examination resolves to ACCEPT (act under audit), ESCALATE (hand the decision to a qualified human), or BLOCK (stop the action before it reaches the patient). The gates are not advisory. They are encoded in a standard operating procedure, the written, version-controlled document that fixes who does what, in what order, and under which thresholds, so that the same action is checked the same way every time and no step depends on an individual's memory at the moment of decision.

The gates fall into three tiers, and Table 2 sets out what each tier checks and what happens when a check fails. The first four gates test numeric thresholds (accuracy, calibration, uncertainty, and tolerance bounds), so that a quantitatively deficient action never advances. Gates five through nine bind the action to external standards and protocol constraints, so that compliance is verified rather than assumed. Gate ten is the catastrophe predicate: if the contemplated action matches a predicate for irreversible harm, the system issues a BLOCK before the patient, regardless of how confident the earlier gates were. This is the hard stop the question demands, and Figure 4 shows it as the final fork between contain and BLOCK.

Safety also depends on what happens after a stop. A BLOCK or an ESCALATE is only useful if a competent human is actually there to receive it, at any hour. The author's adverse-event lineage describes a threefold

humanoid response team configured for 24/7 coverage, so that an escalated case meets a qualified responder rather than a queue [10]. The illustrative H. R. 9510 would make this coverage and the gate suite part of the binding trial record rather than a vendor’s promise [1], so that the existence of the hard stop, and of the team behind it, can be confirmed by an outside party.

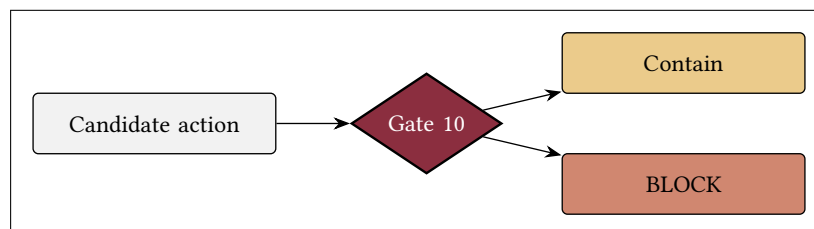


Figure 4. The catastrophe-predicate gate (Mermaid 04 and 07 reproduced as colored TikZ): a candidate action that matches the irreversible-harm predicate is contained or issued a BLOCK before it can reach the patient.

Gate	What it checks	If it fails
Gates 1 to 4 (numeric thresholds)	Accuracy, calibration, uncertainty, and tolerance bounds against pre-set numeric limits	The action is held and ESCALATED to a qualified human for review
Gates 5 to 9 (standards binding)	Conformance to external standards, protocol constraints, and the signed standard operating procedure	The deviation is logged and the action is blocked pending correction
Gate 10 (catastrophe predicate)	Whether the action matches a predicate for irreversible or catastrophic harm	The system issues a BLOCK before the patient, overriding earlier passes

Table 2: The ten gates in three tiers: what each checks and the consequence of a failure, ending in the catastrophe-predicate BLOCK.

A framework cannot promise that nothing will go wrong. It can promise a hard stop before the worst outcomes, and a record of every action that does proceed. The gate, the standard operating procedure behind it, and the 24/7 team that answers an escalation are the concrete answer to the question every clinician carries to the bedside: would I let this system near my own family member, knowing exactly what stops it before it hurts someone?

4 Transparency: can I see why it did that?

A clinician cannot endorse a black box. The third question is whether the system’s reasoning is inspectable and whether a complete, tamper-evident record lets the clinician reconstruct events after the fact. Transparency is the difference between a tool a clinician can defend and one a clinician can only hope was right. The relevant regulatory direction is explicit on this point: the FDA’s action plan for artificial-intelligence and machine-learning software as a medical device places transparency to users and traceability of decisions at the center of trustworthy deployment [11].

The mechanism makes transparency operational rather than rhetorical. Because every action passes through verification before generation and a ten-gate VVUQ examination resolving to ACCEPT, ESCALATE, or BLOCK, each of those steps produces a durable artifact. Together those artifacts form an audit trail: an append-only, hash-chained record in which each entry is cryptographically linked to the one

before it, so that any later alteration breaks the chain and is detectable. The clinician is not asked to trust that the record is complete; the clinician can verify it. The author's records-and-server lineage describes the national MCP servers that hold these records under a common schema [12], and the site documentation package specifies exactly which elements are captured at each step and how they are bound together [13].

Table 3 lists what the record contains and who reads each part, because transparency means different things to different readers. The clinician needs the model's reasoning and the gate results to make and defend a decision; an auditor needs the inputs and the hash chain to confirm that nothing was altered after the fact; a tumor board needs the countersignature that ties a human to the action. Figure 5 shows the same record being written as the action proceeds and then read back by an auditor who reconstructs the event from the record alone.

This is what lets a clinician answer the two questions that decide whether a tool survives contact with real practice. Can I defend this decision at tumor board? Yes, because the model's reasoning and the gate results are on the record, not in anyone's memory. If an auditor asked me tomorrow, could I show exactly what happened and why? Yes, because the inputs, the reasoning, the gate outcomes, and my countersignature are bound into a hash chain that an outside party can replay end to end. A record that can be reconstructed by someone who was not in the room is the test the FDA direction implies and the one this framework is built to pass.

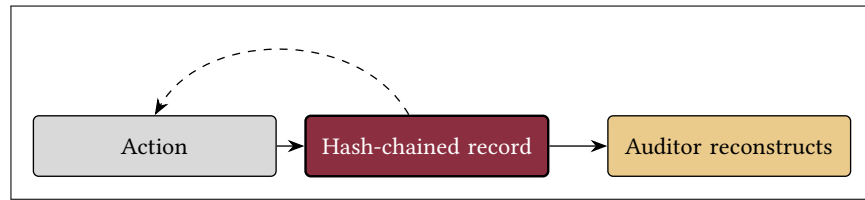


Figure 5. The audit trail (Mermaid 08 reproduced as colored TikZ): each action writes a hash-chained record from which an auditor can later reconstruct exactly what happened and why.

Record element	Contents	Who reads it
Inputs	The patient data, parameters, and context presented to the system	Auditor, sponsor
Model reasoning	The proposed action and the explanation Claude Code generated for it	Clinician, tumor board
Gate results	The pass, ESCALATE, or BLOCK outcome of each of the ten VVUQ gates	Clinician, auditor
Clinician countersignature	The named human who accepted, escalated, or overrode the action	Tumor board, IRB
Hash chain	The cryptographic link binding every entry into a tamper-evident sequence	Auditor, regulator

Table 3: The audit-trail record: what each element contains and which reader relies on it.

Transparency is operational, not rhetorical: the test is whether an auditor can reconstruct exactly what happened, and why, from the record alone. The hash-chained audit trail is that record, and it is the concrete provision that turns a black box into a tool a clinician can defend at tumor board.

5 Oversight: am I still in control?

The clinician must remain the responsible agent. The fourth question asks whether the human stays in control: how to override, pause, or stop the system, and when it escalates a decision to a qualified person

rather than acting alone. The goal is calibrated trust, meaning reliance proportioned to the system's demonstrated capability, which sits between two failure modes that the oversight design must hold apart. Over-trust, or automation bias, is the tendency to defer to a machine even when it is wrong; the clinical-decision-support literature documents how readily it occurs when a confident recommendation appears on screen [3]. Under-trust, or algorithm aversion, is the opposite error of discarding useful guidance. Appropriate reliance is the calibrated middle, and it depends on the human keeping levers that are real rather than ceremonial [2].

The framework provides those levers through two complementary supervision modes. Human-in-the-loop means the clinician sits inside the decision: the system proposes an action under verification before generation, but it cannot act until the clinician accepts it, so a human authorizes each step. Human-on-the-loop means the clinician supervises a running process from above: the system acts under audit within pre-authorized bounds while the clinician monitors and retains the power to intervene at any moment. The ten-gate VVUQ examination decides which mode applies to a given action and, when a proposal falls outside its bounds, takes the ESCALATE path that routes the decision to a qualified human instead of acting alone.

Above both modes sits an unconditional control: the clinician can override and stop the system at any time, for any reason, without satisfying the gates first. Table 4 sets out the four oversight positions, where the clinician sits in each, and what control remains available. Figure 6 shows the same loop, in which a monitored recommendation reaches a decision point that can resolve to execute, escalate, or override. The design intent is that the human is never a spectator: in-the-loop the human authorizes, on-the-loop the human supervises, on escalation the human is summoned, and on override the human simply stops the action.

This is how the framework guards against both automation bias and algorithm aversion at once. The gates and the audit record give the clinician enough evidence to rely on the system when it has earned reliance, which counters aversion; the ever-present override and the escalation path ensure the clinician can refuse or halt it the moment it has not, which counters over-trust. Control, defined this way, is what makes calibrated trust possible rather than merely aspirational.

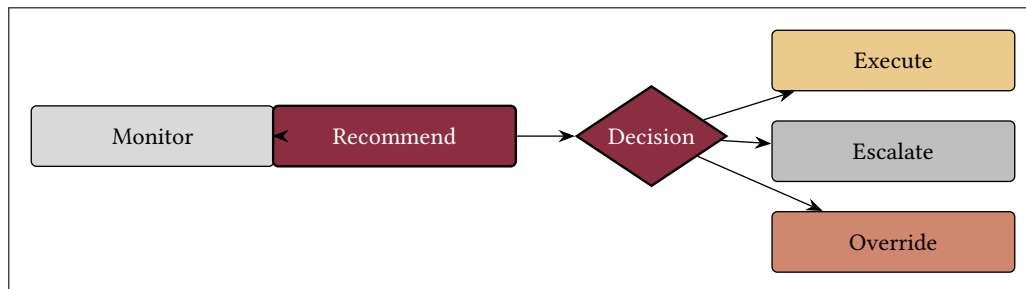


Figure 6. The oversight loop (Mermaid 09 reproduced as colored TikZ): a monitored recommendation reaches a decision that resolves to execute under audit, escalate to a qualified human, or override and stop.

Oversight mode	Where the clinician sits	Control available
Human-in-the-loop	Inside the decision; the action waits for the clinician	Authorize, modify, or refuse each step
Human-on-the-loop	Above a running process within pre-authorized bounds	Monitor in real time and intervene at will
Escalation	Summoned when a proposal exceeds the gates' bounds	Decide the escalated case as a qualified human
Override and stop	Outside the gates, at any moment	Halt the system unconditionally, no gate required

Table 4: The four oversight positions: where the clinician sits and what control each preserves, from human-in-the-loop authorization to an unconditional stop.

Control is not a slogan but a set of switches: a pause, an override, and an escalation path that the clinician can reach at any moment. Those switches, written into the illustrative H. R. 9510 as standing provisions rather than vendor courtesies [1], are the concrete answer to the question of whether the clinician is still in control, and the mechanism that keeps trust calibrated between automation bias and algorithm aversion.

6 Equity: does it work for the patient in front of me?

A clinician treats this patient, not the average patient. The fifth question asks whether performance holds for the specific subgroup in the room, and whether bias is actively surveilled rather than assumed away. The danger is well documented and easy to miss, because a biased system can post excellent overall numbers while failing a particular group. In a widely used health algorithm, reliance on historical cost as a proxy for need systematically understated the illness of Black patients, so that an aggregate that looked fair concealed a large disparity [14]. An average, in other words, is not evidence of equity; it can be a place where inequity hides.

The framework answers this with subgroup performance reporting: the system's accuracy, calibration, and error rates are measured and reported separately for each pre-specified subgroup, not only in aggregate. This reporting is not a one-time audit but a continuous obligation, because case mix and model behavior drift over time. The ten-gate VVUQ examination is configured so that a subgroup in which performance falls below threshold does not silently pass: the action for a patient in an under-performing subgroup is held and routed to the ESCALATE path, and the disparity itself becomes a surveillance priority rather than a footnote.

Table 5 sets out the subgroup axes the framework tracks, the metric reported on each, and the action taken when a disparity appears. The axes span age, ancestry, sex, comorbidity burden, and rural access, because a patient can be disadvantaged on any of them and the relevant axis is rarely the one that is most convenient to measure. Figure 7 shows the structure: overall performance is decomposed by the subgroup audit, which resolves either to equitable performance or to a flagged disparity that enters surveillance.

This is what lets equity be answered by measurement rather than by assurance. The test a clinician can apply is direct: does it work for the patient in front of me, specifically, on the axis that describes this patient? Because performance is reported per subgroup and a shortfall triggers escalation and surveillance, the clinician can ask that question and get a real answer instead of an aggregate that flatters the system. Equity built this way also passes the family test, because the patient in front of the clinician is treated as a member of a measured group, not absorbed into an average that was never about them.

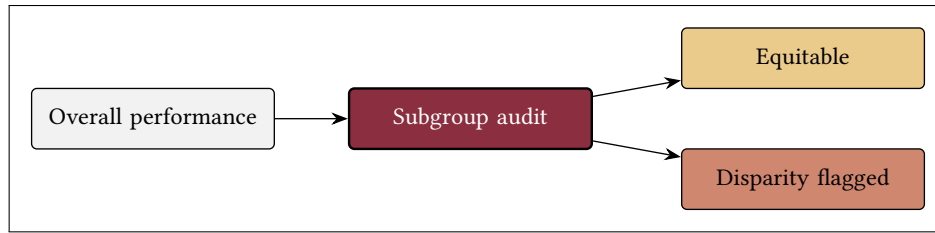


Figure 7. Subgroup equity (Mermaid 10 reproduced as colored TikZ): overall performance is decomposed by a subgroup audit that resolves to equitable performance or to a flagged disparity entering surveillance.

Subgroup axis	Metric reported	Action if disparity
Age	Accuracy and calibration by age band, including older patients	Escalate and add the band to surveillance
Ancestry	Error rates by reported ancestry to detect proxy bias	Escalate and audit the proxy variables
Sex	Sensitivity and specificity reported by sex	Escalate and recheck the training balance
Comorbidity burden	Performance stratified by comorbidity load	Escalate; widen the bound for complex cases
Rural access	Outcome and access gaps for remote sites	Escalate and route to the 24/7 response team

Table 5: Subgroup audit axes: the metric reported on each axis and the action taken when a disparity is detected.

Equity is answered by measurement: performance reported per subgroup, with disparities surfaced as surveillance priorities rather than buried in an average. Writing per-subgroup reporting and bias surveillance into the illustrative H. R. 9510 as a standing requirement [1] is the concrete provision that lets a clinician trust the system works for the patient in front of them, and not merely for the patient on average.

7 Reliability: will it be the same tomorrow, at 3 a.m.?

The clinician needs the same answer tomorrow as today. The sixth question asks whether the system behaves consistently over time and load, how model drift is detected and handled, and how uncertainty is quantified and communicated. Reliability is not the same as accuracy: a system can be accurate on the day it is validated and quietly degrade as the population, the instruments, or the model itself shift. The engineering discipline for this is a formal credibility assessment, and the framework follows the structure of ASME V&V 40, which grades how much trust a computational model has earned for a stated use and ties that trust to the consequences of being wrong [15].

The mechanism turns that discipline into running controls. Uncertainty quantification, the third leg of the ten-gate VVUQ examination, attaches a confidence bound to every estimate, and the gate compares that bound against a pre-set tolerance before the action proceeds: an estimate whose uncertainty is too wide does not pass quietly, it takes the ESCALATE path. Around that point-of- decision check sit three further controls: a reproducibility check that confirms the same input still yields the same output, a drift monitor that watches inputs and performance for a statistically significant shift, and a scheduled revalidation that re-earns credibility on a fixed cadence. Together these implement the adoption action of continuous monitoring, so reliability is maintained rather than assumed after go-live.

Table 6 sets out each control, the trigger that fires it, and the response it produces. Figure 8 shows the point-of-decision case: an estimate carries a confidence interval to the gate, which asks whether the estimate sits within bound and resolves to report or escalate accordingly. The author’s verification-automation lineage demonstrates this at scale, where a code-generation process was held to account against the FDA FAERS adverse-event record rather than against its own internal claims [16], which is the external check that keeps reproducibility honest.

The phrase “at 3 a.m.” is the real test, because reliability that holds only during business hours is not reliability. The night-shift case must be treated exactly like noon: the same gates, the same uncertainty bounds, the same drift monitor, and the same escalation to a qualified human if a bound is exceeded. The illustrative H. R. 9510 would make scheduled revalidation and drift monitoring standing obligations of the trial rather than optional maintenance [1], so that a clinician arriving at 3 a.m. inherits the same earned credibility as the team that signed off at noon.

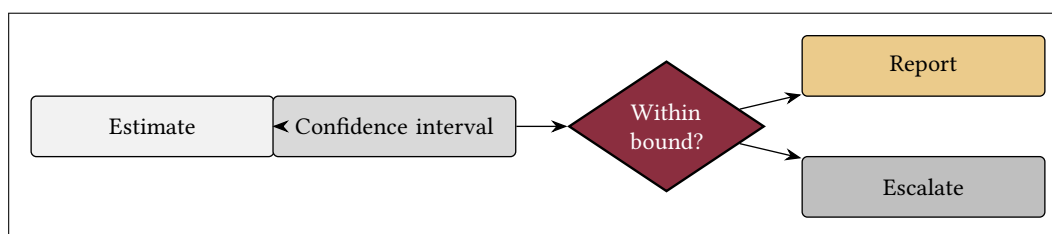


Figure 8. The uncertainty gate (Mermaid 15 reproduced as colored TikZ): an estimate carries a confidence interval to the gate, which reports the result if it is within bound and escalates to a qualified human if it is not.

Control	Trigger	Response
Reproducibility check	The same input yields a different output than recorded	Halt and investigate before further use
Drift monitor	Input distribution or performance shifts beyond a set threshold	Flag the drift and open continuous monitoring
Uncertainty bound	An estimate’s confidence interval exceeds the gate tolerance	Escalate the case to a qualified human
Scheduled revalidation	The fixed cadence or a triggering drift event is reached	Re-run the credibility assessment and gates

Table 6: Reliability controls: the trigger that fires each control and the response it produces, implementing continuous monitoring after go-live.

Reliability is answered by honesty about doubt: a quantified confidence interval, a plan for drift, and a guarantee that 3 a.m. is treated exactly like noon. The uncertainty gate, the drift monitor, and scheduled revalidation are the concrete controls that let a clinician trust the system will give the same defensible answer tomorrow, on the night shift, as it gave the day it was validated.

8 Accountability: who is responsible when it is wrong?

A clinician must know where responsibility sits. The seventh question asks who answers when something goes wrong, what is documented and by whom, and how liability is allocated across the human and the system. A diffuse answer is no answer: if everyone is responsible, no one is, and the clinician is left

holding a risk that was never theirs alone to carry. Accountability in this framework is therefore drawn explicitly, with each role named and each obligation written down before the trial begins.

The instrument that draws it is the standard operating procedure, the written, signed, version-controlled document that fixes who does what, under which thresholds, and in what order. Because the standard operating procedure is signed by named parties, it converts an abstract division of labor into a record of who agreed to what. Sitting alongside it is the audit trail, the append-only, hash-chained record in which each action, gate result, and countersignature is cryptographically linked, so that any action can be traced from outcome back to the human or system responsible. The author's site documentation package specifies the contents and custody of these records [13], and the regulatory-adaption lineage maps the roles onto existing frameworks: the adaption of 21 CFR Part 312 places sponsor and investigator duties on a familiar legal footing [17], while the adaption of the ICH harmonised guideline aligns the same roles with international good clinical practice [18].

Table 7 sets out, for each party, what that party is responsible for and where the responsibility is documented. The clinician owns the decision to accept, escalate, or override, recorded by countersignature in the audit trail. The sponsor owns the protocol and oversight, the developer owns the verified behavior of the system, the IRB owns the ethical approval and its conditions, and the auditor owns independent confirmation that the record matches what occurred. Figure 9 shows the same structure converging on the signed standard operating procedure, with the auditor positioned to read across all of it.

This is what lets a clinician answer the question that matters when an action is later challenged. If an auditor asked me tomorrow, could I show exactly what happened and who was responsible? Yes, because the standard operating procedure names the responsible party for each function and the audit trail binds each action to that party. Accountability drawn this way also protects the clinician, because it makes visible the obligations that belong to the sponsor and the developer rather than letting them settle silently onto the person at the bedside.

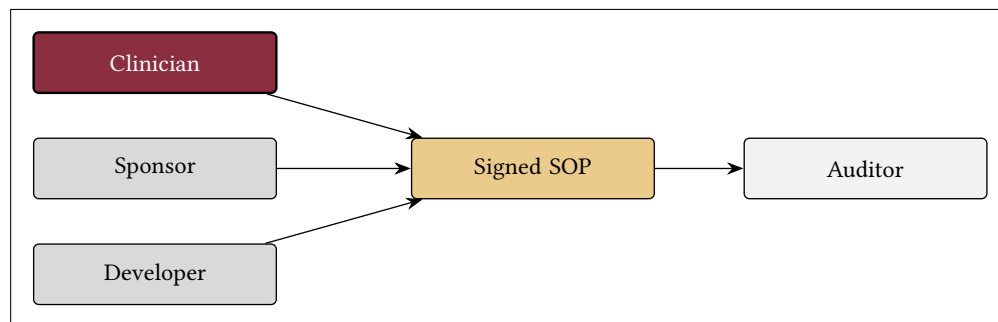


Figure 9. Where accountability sits (Mermaid 13 reproduced as colored TikZ): clinician, sponsor, and developer obligations converge on a signed standard operating procedure that the auditor reads across.

Party	Responsible for	Documented in
Clinician	The decision to accept, escalate, or override an action	Countersignature in the audit trail
Sponsor	The protocol, oversight, and adverse-event response	Signed standard operating procedure
Developer	The verified behavior of the system and its gates	Verification record and gate results
IRB	Ethical approval and the conditions attached to it	Approval letter and standard operating procedure
Auditor	Independent confirmation that the record matches events	Audit report against the hash chain

Table 7: Responsibility allocation: what each party is responsible for and where that responsibility is recorded, anchored by the signed standard operating procedure and the audit trail.

Accountability is answered by drawing it: named roles, a signed procedure, and a documentation chain that an auditor can follow from action to owner. The signed standard operating procedure and the hash-chained audit trail are the concrete mechanisms that tell a clinician, before the trial starts, exactly who is responsible when something is wrong.

9 Workflow: does it fit how I actually work?

The best safe tool fails if it does not fit. The eighth question asks whether the system reduces burden or adds to it, and how it changes consent and the clinic day, because a tool that fights the workflow will be worked around or abandoned. This is not a soft concern. Clinicians discard even useful guidance once it has disappointed them, and they do so more readily than they discard a fallible human colleague; the phenomenon of algorithm aversion shows that confidence, once lost to a visible error or to daily friction, is hard to recover [4]. A system that adds clicks, interrupts the consult, or duplicates documentation will lose the clinician's trust regardless of how well its gates perform.

The answer is workflow integration: the system is embedded into the existing steps of the clinic day rather than bolted alongside them. Each action still passes through verification before generation and the ten-gate VVUQ examination, but the clinician experiences this as a recommendation that arrives in place, with its reasoning and an ACCEPT, ESCALATE, or BLOCK already attached, and a countersignature that doubles as the documentation. Consent is handled the same way, as a verified process woven into the encounter rather than a separate hurdle; the role of an honest, well-supported consent conversation in earning patient trust is well established [19], and the author's patient-journey lineage traces a single patient moving from referral through consent to follow-up without the workflow ever leaving the clinician's normal path [20].

Table 8 sets out the clinic touchpoints and contrasts each one without the system and with it, so the clinician can see where burden is removed rather than added. Referral and eligibility, informed consent, the clinic decision, documentation, and follow-up monitoring each become a step where the verified record is produced as a by-product of the work rather than as extra work. Follow-up in particular is where continuous monitoring lives: the same drift and uncertainty controls that protect reliability run quietly after the visit, surfacing only when a patient needs attention.

Figure 10 places this single clinic encounter in its larger arc, from one bedside decision to site-level practice to trial-wide calibrated trust, meaning reliance proportioned to demonstrated capability. The author's national-platform work shows how that arc scales beyond one clinic while the workflow at each

bedside stays recognizable [21]. The point of the scaling is not uniformity for its own sake but that the same verified, low-friction experience a clinician trusts at one bedside is the experience an entire trial can be built on.

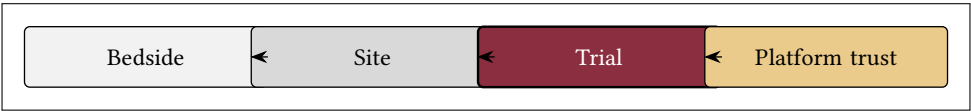


Figure 10. From bedside to trial-wide trust (Mermaid 20 reproduced as colored TikZ): a single verified bedside decision scales through the site and the trial to platform-wide calibrated trust.

Workflow touchpoint	Without the system	With the system
Referral and eligibility	Manual chart review against criteria, easily delayed	Eligibility checked and recorded as the referral arrives
Informed consent	A separate hurdle bolted onto the visit	A verified consent process woven into the encounter
The clinic decision	Judgment made and documented from memory	A recommendation in place with reasoning and a gate result
Documentation	Extra notes written after the fact	A countersignature that is itself the audit record
Follow-up monitoring	Periodic, manual, easy to miss	Continuous monitoring that surfaces only when needed

Table 8: The clinic day at each workflow touchpoint, contrasted without and with the system, showing where workflow integration removes burden rather than adding it.

Workflow is answered by fit: a system embedded in the clinic day and the consent process, measured for burden, and scaled from one bedside decision to trial-wide calibrated trust. Workflow integration and continuous monitoring are the concrete provisions that make the system something a clinician uses rather than works around.

The eight questions are not eight separate tests but eight doors into the same room. Each one opens onto a specific mechanism, a gate, a record, a role, or a control, and the clinician who has walked through all eight has the evidence to calibrate trust rather than to grant or withhold it wholesale. Table 9 gathers the eight questions and the mechanism that answers each, so the framework can be read at a glance and carried into a credentialing committee or a tumor board.

Question	The mechanism that answers it
Competence	Local validation and an assurance run, with 172 of 172 gate tests passed before action
Safety	The ten-gate VVUQ examination and the catastrophe-predicate BLOCK before the patient
Transparency	Explainable output and a complete, hash-chained audit trail
Oversight	Human-in-the-loop control, the ESCALATE path, and an unconditional override
Equity	Subgroup performance reporting and continuous bias surveillance
Reliability	Reproducibility checks, drift monitoring, and quantified uncertainty
Accountability	Named roles and a signed standard operating procedure
Workflow	Integration into the clinic day and consent, with continuous monitoring

Table 9: The eight confidence questions and the specific mechanism that answers each.

A clinician who can answer all eight questions has what calibrated trust requires: not faith, but the evidence to rely on the system exactly as far as it has earned, and no further. The three questions that opened this framework now resolve into one. Would I let this system near my own family member, could I defend the decision at tumor board, and could I show an auditor exactly what happened and why? Each can be answered yes, because a gate, a record, a role, or a control stands behind it. That is the adoption decision this framework asks the clinician to make, and to sign.

10 Appendix A: figure compendium

This appendix reproduces the remaining figures from the colored Mermaid catalog (adoption/mermaid/) as colored TikZ, so the framework carries the full set of exhibits in the strict palette. Each caption names the Mermaid source.

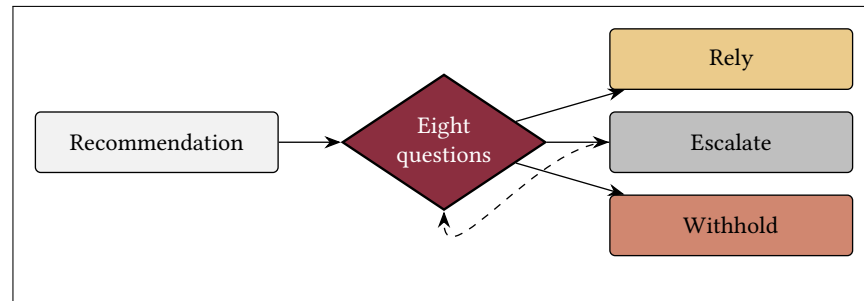


Figure 11. The bedside trust decision (Mermaid 01): the eight questions gate one decision with three guarded outcomes.

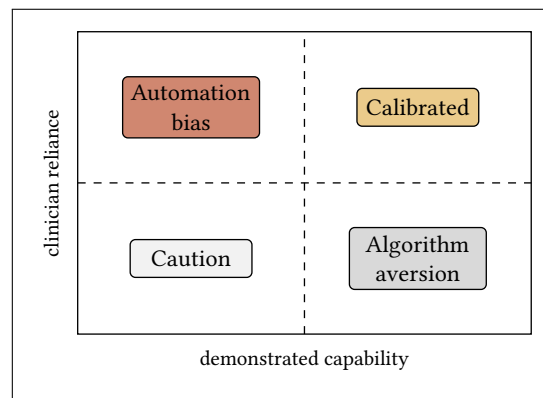


Figure 12. The calibrated-trust band (Mermaid 05): calibrated trust sits between automation bias and algorithm aversion.

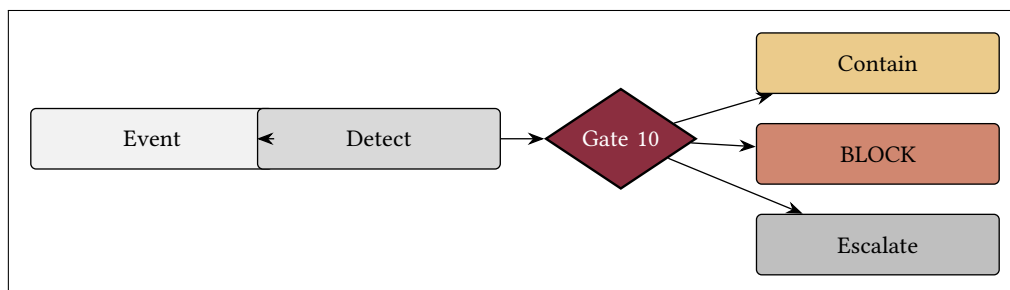


Figure 13. Safety containment (Mermaid 07): a detected event is gated into containment, a hard block, or escalation.

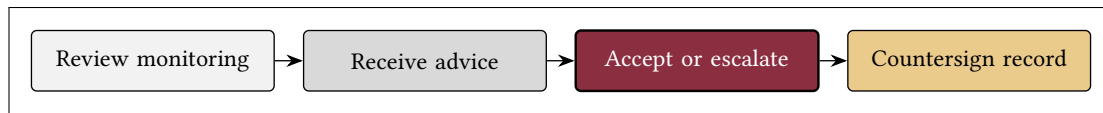


Figure 14. A clinician's day (Mermaid 11): the system enters the clinic day at four scored touchpoints.

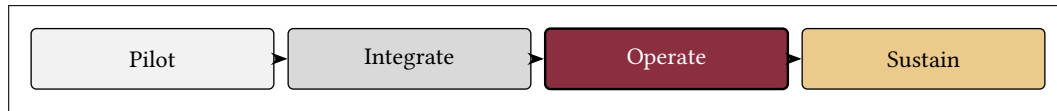


Figure 15. The adoption maturity ladder (Mermaid 12): four phases from pilot to standard work.

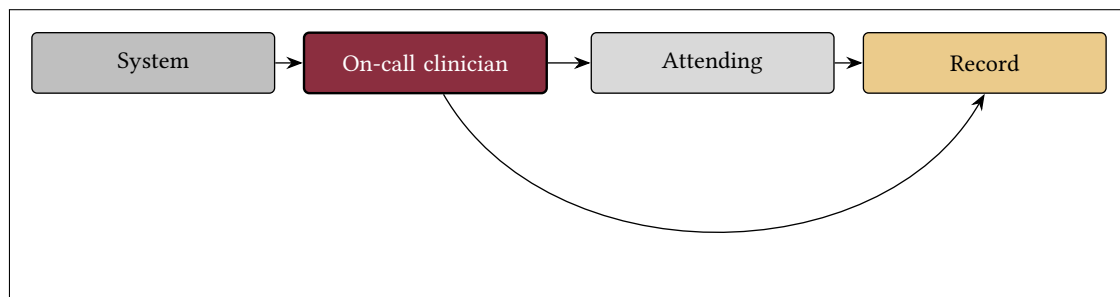


Figure 16. The escalation pathway (Mermaid 14): an uncertain gate escalates, and every step writes to the record.

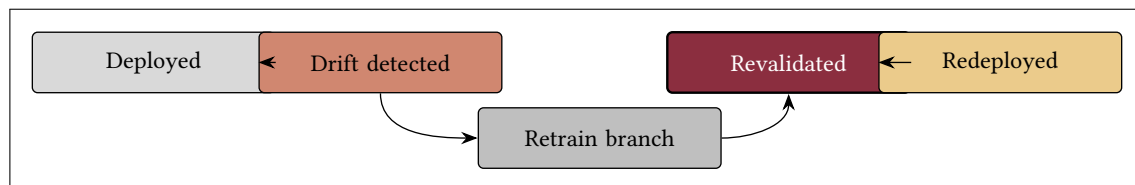


Figure 17. Model revalidation (Mermaid 16): a retrain branch revalidates against the gates before it merges back into service.

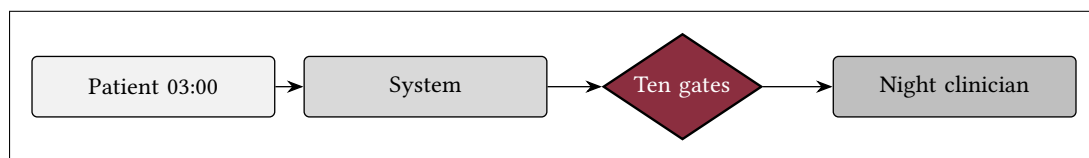


Figure 18. The night-shift scenario (Mermaid 17): 03:00 is treated exactly like noon, by the same gates.

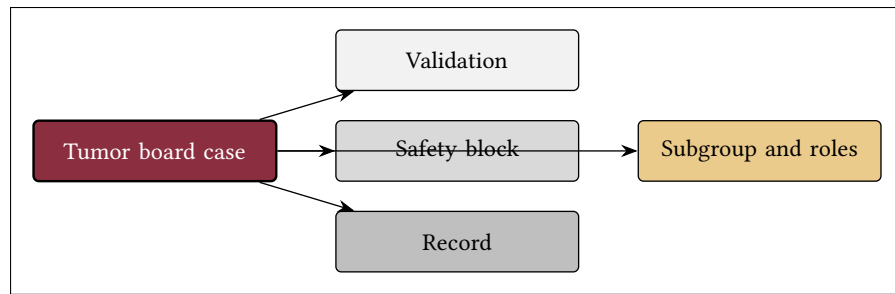


Figure 19. Defensible at tumor board (Mermaid 18): the parallel supports a clinician brings to peer review.

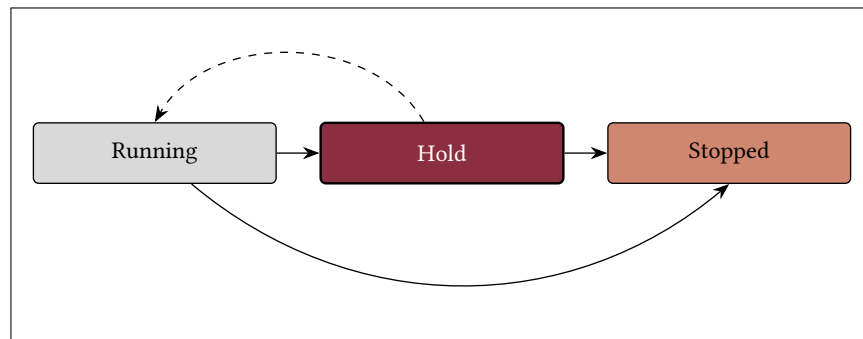


Figure 20. Override and the stop authority (Mermaid 19): a clinician can pause and resume, or override to a full stop, and the predicate can stop it alone.

Taken together, the twenty figures trace one claim from a single bedside decision (Figure 11) to trial-wide and platform trust (Figure 10), and each exhibit answers one of the eight questions in the strict palette. None of them asks the clinician to trust more than the evidence warrants; each shows where a gate, a record, a role, or a control makes that trust calibrated.

References

- [1] K. Kawchak, “H. r. 9510 (bill v5.0) 2026,” jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [2] J. D. Lee and K. A. See, “Trust in automation: Designing for appropriate reliance,” *Human Factors*, vol. 46, no. 1, pp. 50–80, 2004. https://doi.org/10.1518/hfes.46.1.50_30392.
- [3] K. Goddard, A. Roudsari, and J. C. Wyatt, “Automation bias: a systematic review of frequency, effect mediators, and mitigators,” *Journal of the American Medical Informatics Association*, vol. 19, no. 1, pp. 121–127, 2012. <https://doi.org/10.1136/amiajnl-2011-000089>.
- [4] B. J. Dietvorst, J. P. Simmons, and C. Massey, “Algorithm aversion: People erroneously avoid algorithms after seeing them err,” *Journal of Experimental Psychology: General*, vol. 144, no. 1, pp. 114–126, 2015. <https://doi.org/10.1037/xge0000033>.
- [5] E. J. Topol, “High-performance medicine: the convergence of human and artificial intelligence,” *Nature Medicine*, vol. 25, no. 1, pp. 44–56, 2019. <https://doi.org/10.1038/s41591-018-0300-7>.
- [6] K. Kawchak, “Mobile pancreatic cancer unitree h2 surgical humanoid with priority vvuq,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [7] K. Kawchak, “Vvuq oncology clinical trial llm verification automation priority over existing generated code,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [8] L. T. Kohn, J. M. Corrigan, and M. S. Donaldson, eds., *To Err Is Human: Building a Safer Health System*. Washington, DC: National Academies Press, 2000. Institute of Medicine. <https://doi.org/10.17226/9728>.
- [9] M. A. Makary and M. Daniel, “Medical error - the third leading cause of death in the us,” *BMJ*, vol. 353, p. i2139, 2016. <https://doi.org/10.1136/bmj.i2139>.
- [10] K. Kawchak, “Threefold humanoid 24/7 adverse event oncology trial response team: 4-site rotation,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [11] U.S. Food and Drug Administration, “Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) Action Plan.” U.S. Food and Drug Administration, 2021. <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device>.
- [12] K. Kawchak, “National mcp servers for physical ai oncology clinical trial systems,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [13] K. Kawchak, “Physical ai oncology clinical trial site complete documentation package,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [14] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science*, vol. 366, no. 6464, pp. 447–453, 2019. <https://doi.org/10.1126/science.aax2342>.

- [15] American Society of Mechanical Engineers, “Assessing Credibility of Computational Modeling Through Verification and Validation: Application to Medical Devices (ASME V&V 40-2018),” 2018. <https://www.asme.org/codes-standards/find-codes-standards/v-v-40-assessing-credibility-computational-modeling-verification-validation>
- [16] K. Kawchak, “Code generation competition: 16 proprietary vs. open-source llms & iterative learning based on fda adverse event reporting system,” dec 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [17] K. Kawchak, “Adaption: 21 cfr part 312, end-to-end physical ai oncology clinical trial unification,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [18] K. Kawchak, “Adaption: Ich harmonised guideline, end-to-end physical ai oncology clinical trial unification,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [19] L. L. Novak, S. George, K. A. Wallston, Y. A. Joosten, T. L. Israel, C. L. Simpson, and C. H. Wilkins, “Patient stories can make a difference in patient-centered research design,” *Journal of Patient Experience*, vol. 7, no. 6, pp. 1438–1444, 2020. <https://doi.org/10.1177/2374373520958340>.
- [20] K. Kawchak, “A cancer patient’s journey through a regulated physical ai oncology trial,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [21] K. Kawchak, “National platform for physical ai oncology trials,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [22] A. Ingber, *Regulating Emotion AI in the United States: Insights from Empirical Inquiry*. PhD thesis, Deep Blue Repositories, 2025. Working paper / Doctoral dissertation.
- [23] S. Moreland-Russell, C. Barbero, S. Andersen, N. Geary, E. A. Dodson, and R. C. Brownson, ““Hearing from all sides”: How legislative testimony influences state level policy-makers in the United States,” *International Journal of Health Policy and Management*, vol. 4, no. 2, pp. 91–98, 2015. <https://doi.org/10.15171/ijhpm.2015.13>.
- [24] M. Putturaj, ““Honouring the storyteller”: The potential of Playback Theatre in health policy and systems research,” *Health Policy and Planning*, vol. 40, no. 7, pp. 809–817, 2025. <https://doi.org/10.1093/heapol/czad000>.
- [25] G. Kok, L. K. Bartholomew, G. S. Parcel, N. H. Gottlieb, and M. E. Fernández, “Finding theory- and evidence-based alternatives to fear appeals: Intervention mapping,” *International Journal of Psychology*, vol. 49, no. 2, pp. 98–107, 2013. <https://doi.org/10.1002/ijop.12001>.
- [26] K. Kawchak, “H. r. 9510 (bill v4.0) 2026,” jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [27] K. Kawchak, “H. r. 9510 (bill v3.0) 2026,” jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [28] K. Kawchak, “Verification before generation act of 2026,” jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [29] K. Kawchak, “Vvuq physical ai oncology trial bill,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

- [30] K. Kawchak, “2030: 60 second pancreatic ductal adenocarcinoma robotic whipple procedure & daraxonrasib simulation,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [31] K. Kawchak, “2030: 60 second glioblastoma ai robotic surgery,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [32] K. Kawchak, “Patient priority of proposed u.s. bills for physical ai oncology clinical trials,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [33] K. Kawchak, “Accelerated patient prediction in physical ai oncology clinical trials: 4 extensive simulations,” may 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [34] K. Kawchak, “Fully automated sponsor: Physical ai oncology clinical trial platform,” apr 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [35] K. Kawchak, “Adaption: 21 cfr part 50, end-to-end physical ai oncology clinical trial unification,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [36] K. Kawchak, “Trialmcp: Mcp servers for physical ai oncology clinical trial systems,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [37] K. Kawchak, “Federated learning physical ai oncology trials unification,” feb 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [38] K. Kawchak, “Unification standard level for physical ai oncology trials,” feb 2026. ORCID record work; source snapshot exported from ORCID webarchive.
- [39] K. Kawchak, “Ai peer review acceleration of llm-generated glioblastoma clinical trial patient matching ml, fda/ich/iso, and fastapi,” dec 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [40] K. Kawchak, “Llm-generated glioblastoma drug synergy machine learning: From rapid code prototypes to project deliverables package,” nov 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [41] K. Kawchak, “End-to-end oncology clinical trial llm efficiency for industry adoption with fda/ich regulations,” oct 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [42] K. Kawchak, “Accelerating fda compliance and cost efficiency of in silico clinical trials via ai digital twin pancreatic cancer simulation,” oct 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [43] K. Kawchak, “Qsp metastatic pancreatic cancer ai clinical trial simulation from protocol to prediction: Code, vvuq, and playbook,” aug 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [44] K. Kawchak, “Chatgpt 100,000 patient 24-month in silico phase iii 5-arm pancreatic cancer clinical trial triplicate,” jul 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [45] K. Kawchak, “End-to-end pancreatic ductal adenocarcinoma digital twin clinical trial proposals,” jun 2025. ORCID record work; source snapshot exported from ORCID webarchive.

- [46] K. Kawchak, “10 year glioblastoma clinical trial meta-analyses by autonomous ai at scale. survival, hr, ae, and rob scored in ai reports and charts, including verifications,” may 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [47] K. Kawchak, “Ai revolution toward the cure of lung adenocarcinoma,” apr 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [48] K. Kawchak, “Autonomous llm agent and scalable reasoning llm for generating cancer drug industry cost solutions,” mar 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [49] K. Kawchak, “Cost containment of global monoclonal antibody drugs and cancer clinical trials via llm focused reasoning,” feb 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [50] K. Kawchak, “Clinical decision support based on bevacizumab cancer trials and pushing the limitations of advanced llms,” jan 2025. ORCID record work; source snapshot exported from ORCID webarchive.
- [51] K. Kawchak, “Cancer vs. conversational artificial intelligence,” dec 2024. ORCID record work; source snapshot exported from ORCID webarchive.
- [52] K. Kawchak, “mab bioprocess engineering in-context table forecasts using conversational ai literature insight generations,” dec 2024. ORCID record work; source snapshot exported from ORCID webarchive.
- [53] K. Kawchak, “Monoclonal antibody bioprocess engineering advancements using conversational artificial intelligence,” oct 2024. ORCID record work; source snapshot exported from ORCID webarchive.
- [54] K. Kawchak, “Paclitaxel biosynthesis ai breakthrough,” oct 2024. ORCID record work; source snapshot exported from ORCID webarchive.
- [55] K. Kawchak, “High dimensional and complex spectrometric data analysis of an organic compound using large multimodal models and chained outputs,” sep 2024. ORCID record work; source snapshot exported from ORCID webarchive.
- [56] K. Kawchak, “Lmm spectrometric determination of an organic compound,” aug 2024. ORCID record work; source snapshot exported from ORCID webarchive.
- [57] K. Kawchak, “Lmm chemical research with document retrieval,” aug 2024. ORCID record work; source snapshot exported from ORCID webarchive.

Acknowledgments

The author acknowledges Anthropic for access to Claude Code for the instruction, drafting, and assembly of this framework; OpenAI for ChatGPT subject support; and Google Gemini for additional research assistance.

Ethical Disclosures

The author declares no competing interests. This is an independent framework and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. H. R. 9510 is an illustrative bill number; it is not enacted law and this framework is not legal or clinical advice. All supporting numbers are simulation or illustrative results unless tied to a cited source.

Rights and Permissions

This framework is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0). To view a copy, visit <https://creativecommons.org/licenses/by/4.0/>. Reproduced public-domain U.S. Government text is used under 17 U.S.C. § 105.

Cite This Framework

Kawchak K. Earning the Clinician's Trust: An Eight-Question Confidence Framework for Verified Physical AI in Oncology Trials. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

Data Availability

All sources, figures, and prompt records supporting this framework are openly available in the `kevinkawchak/law-physical-ai-trials` repository under `adoption/` (the prompt, the four sub-prompts, the colored Mermaid catalog, and the three framework stages), and build on the prior review under `review/` and the author's Zenodo deposits. All data sets are synthetic or illustrative; no patient-identifiable information is included.